

Advanced (Habanero)

Q1. Solve the inequality $2x + 5 > 11$

- (A) $x > 3$
- (B) $x < 3$
- (C) $x \geq 3$
- (D) $x \leq 3$
- (E) $x = 3$

Q2. Graph the solution to $x - 2 < 7$ on a number line

- (A) $(-\infty, 9)$
- (B) $(-\infty, 5)$
- (C) $(-5, \infty)$
- (D) $(-9, \infty)$
- (E) $[5, \infty)$

Q3. Write the solution to $x + 1 > 4$ in interval notation

- (A) $(3, \infty)$
- (B) $(-\infty, 3)$
- (C) $[3, \infty)$
- (D) $(-3, \infty)$
- (E) $(-\infty, -3)$

Q4. Solve the inequality $-3x < 12$

- (A) $x > -4$
- (B) $x < -4$
- (C) $x \geq -4$
- (D) $x \leq -4$
- (E) $x = -4$

Q5. Solve the compound inequality $-2 < x < 5$

- (A) $(-2, 5)$
- (B) $[-2, 5]$
- (C) $(-\infty, -2) \cup (5, \infty)$
- (D) $(-\infty, -2] \cup [5, \infty)$
- (E) $(-2, \infty)$

Q6. Solve the inequality $x^2 - 4x - 3 > 0$

- (A) $(-1, 3)$
- (B) $(-\infty, -1) \cup (3, \infty)$
- (C) $(-1, 3)$
- (D) $(-\infty, -3] \cup [1, \infty)$
- (E) $(-3, 1)$

Q7. Solve the inequality $\frac{x+2}{x-2} > 0$

- (A) $(-\infty, -2) \cup (2, \infty)$
- (B) $(-2, 2)$
- (C) $(-\infty, -2] \cup [2, \infty)$
- (D) $[-2, 2)$
- (E) $(-2, 2)$

Q8. Solve the inequality $2x^2 + 5x - 3 < 0$

- (A) $(-3, 1)$
- (B) $(-\infty, -3) \cup (1, \infty)$
- (C) $(-3, 1)$
- (D) $(-1, 3)$
- (E) $(-\infty, -1) \cup (3, \infty)$

Open Ended Questions (FRQ)

Q9. Solve the inequality $x^2 - 5x + 6 > 0$ and graph the solution on a number line. Write the solution in interval notation.

Q10. Solve the system of inequalities $x + y > 2$, $x - y < 1$, and $y > -1$. Graph the solution on a coordinate plane.

Chef's Tips

- Make sure to read the questions carefully before answering
- Use a number line to help visualize the solutions to the inequalities
- Check your work by plugging in test points into the original inequality